Local Shocks, Global Ripples: Climate Disruptions in Maritime Trade Networks

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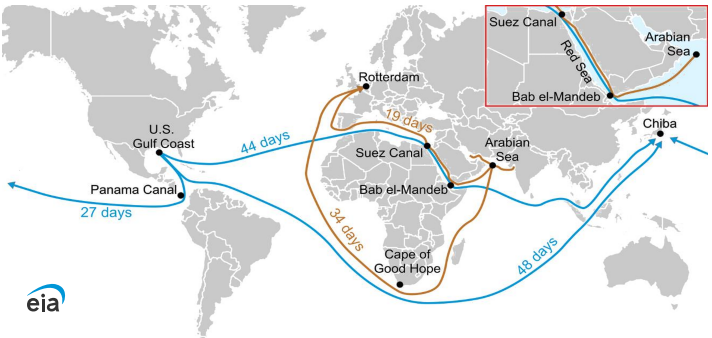


Trade and Climate Adaptation

- Climate shocks affect productivity, economic growth, and the geography of economic activity [Dell et al., 2012, Cruz and Rossi-Hansberg, 2024].
- Trade can moderate climate damages by allowing economies to adjust sourcing and production across locations and sectors [Gouel and Laborde, 2021, Rudik et al., 2022, Nath, 2025].

- But these adjustments depend on transportation networks: around 80% of international trade in goods, by volume, travels by sea [UNCTAD, 2024].
- Climate shocks can also disrupt shipping routes and ports, transmitting economic losses through trade networks:
  - Arctic route openings [Bekkers et al., 2018]
  - Sea-level threats to ports [Hanson and Nicholls, 2020]
  - Weather disruptions [Dunbar et al., 2023, Feng et al., 2025].

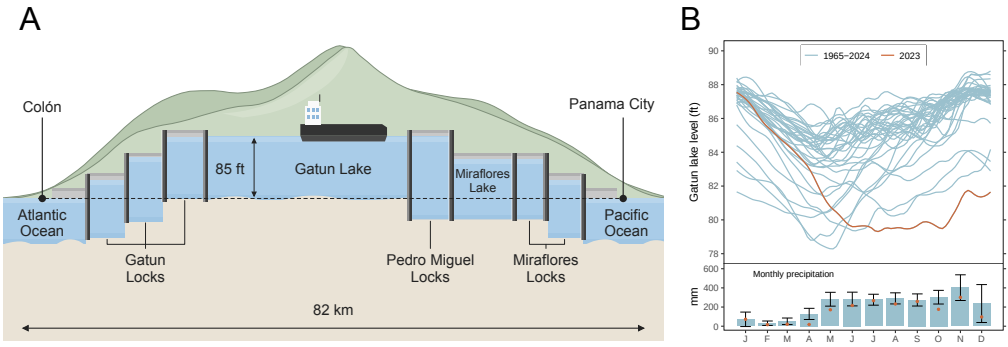
Chokepoints and the Reordering of Global Trade



Source: U.S. Energy Information Administration (2024)

- Global trade routes depend on maritime chokepoints [Känzig and Raghavan, 2025]
- Most disruptions are geopolitical or accidental—but **the Panama Canal capacity is governed by freshwater availability** in Gatun Lake

The Panama Canal Hub



- Operates via a **lock system** fed by Gatun Lake—link between hydrology and capacity
- **2023–2024 Drought** ⇒ FY2024 transits ↓ **29.3%** (vs. FY2023) [PCA, 2026].
- Handles ∼5% of world trade; 15% of U.S. waterborne foreign trade [BTS, 2024]

Research question

How does a climate-induced reduction in shipping capacity at a critical transportation node propagate through global trade networks?

I trace its effects through four stages:

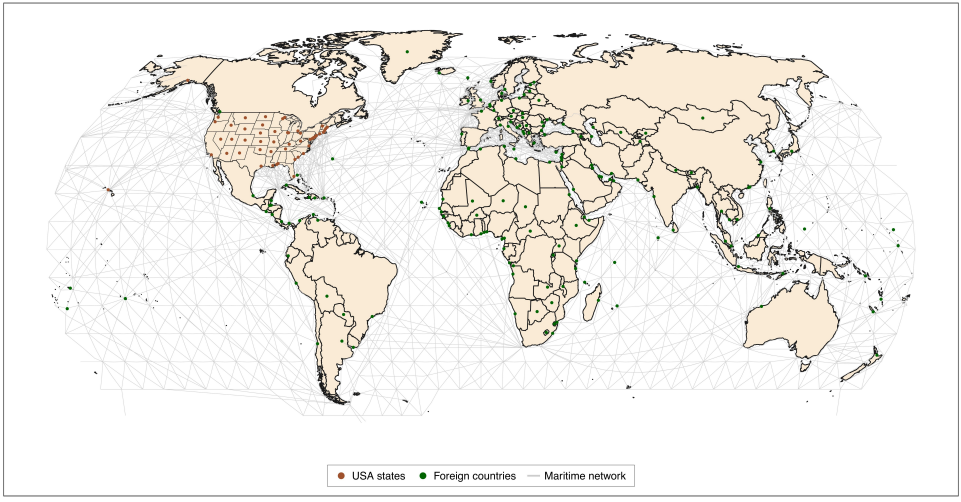
- **Canal capacity:** Estimate how low-water exposure affects Canal operations.
- **Trade and rerouting:** Estimate trade responses to low water by Canal dependence and document rerouting.
- **Network propagation:** Use a network trade model to quantify changes in trade costs and market access.
- **Welfare incidence:** Quantify welfare effects across economies.

Data Sources

- **Canal Operations**†: Panama Canal Authority (1990–2025) ▸ Detail
  - Transit statistics (2003–2025): transits, booked/auctioned slots, toll revenue, net tonnage
  - Hydrological & Meteorological‡: lake levels, inflows, runoff precipitation, temperature
- **Trade Flows**†: U.S. Census Bureau & UN Comtrade (2010–2025) ▸ Detail
  - U.S. imports and exports by state and commodity.
  - Bilateral trade by product and partner: 166 countries
- **Maritime Network**: IMF Port Watch & Eurostat–Marnet ▸ Detail
  - Representative ports and global sea routes [Shen et al., 2020]
  - ∼14,000-node routing graph — least-cost paths via Dijkstra algorithm
- **AIS Vessel Movements**‡: Global Fishing Watch (2012–2026) ▸ Detail
  - Global cargo and tanker port visits and vessel identities
  - ∼11 million harmonized port-to-port movements across ∼30,000 vessels

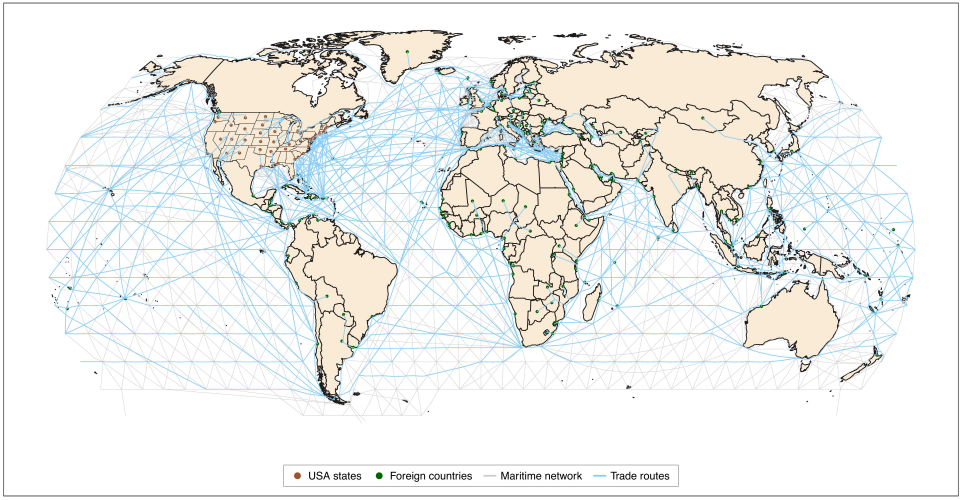
†: Monthly frequency, ‡: Daily frequency.

Maritime Trade Routes



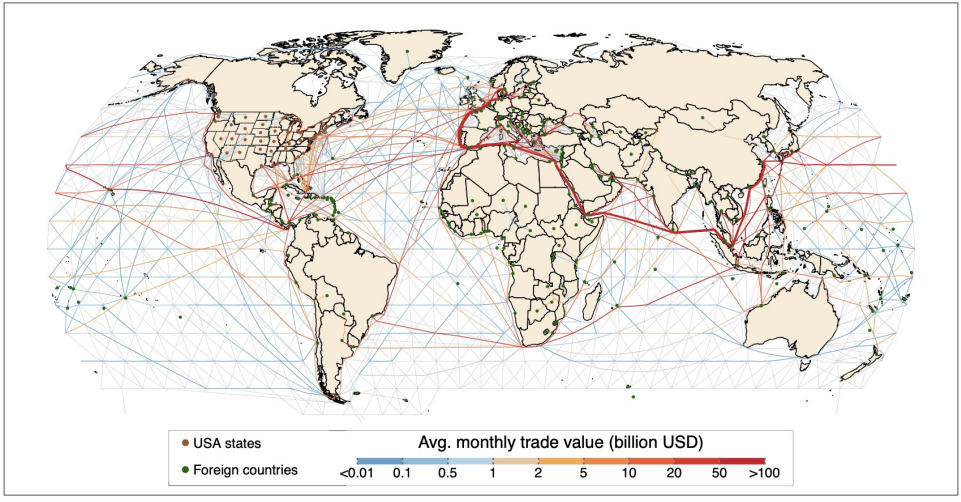
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Maritime Trade Routes



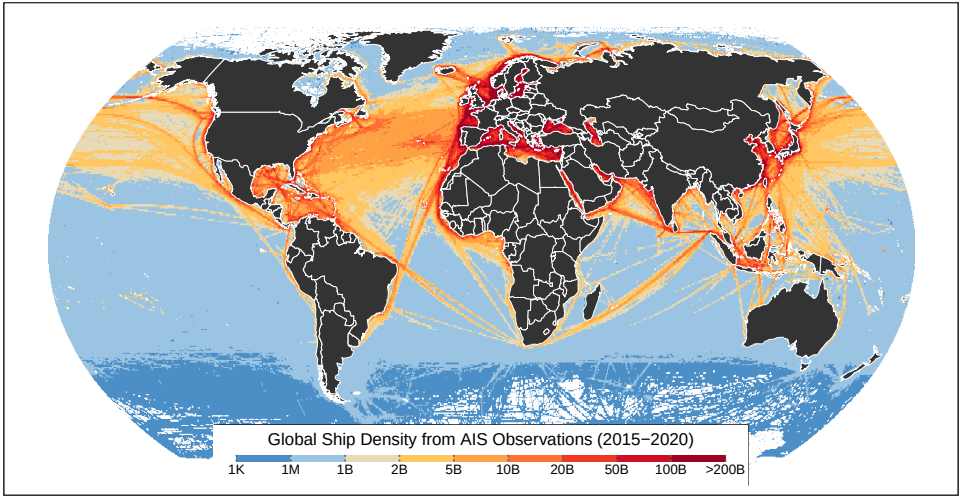
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Maritime Trade Routes



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Global Shipping Traffic Density



Source: World Bank

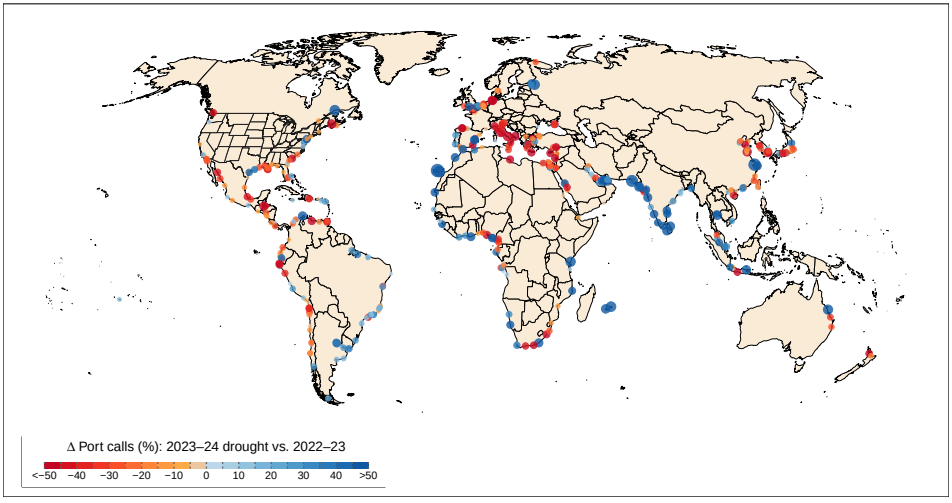
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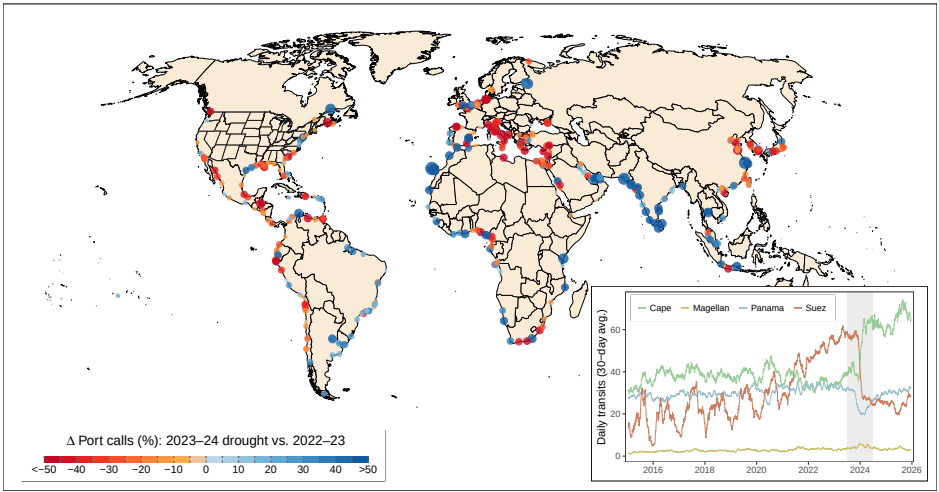
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AIS Vessel Movements



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AIS Vessel Movements

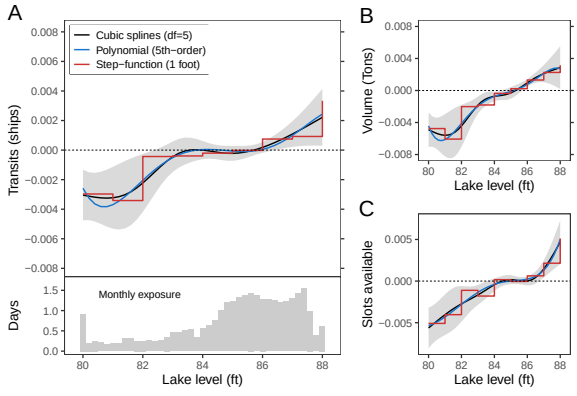


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Lake-Level Exposure and Canal Operations

$$y_t = \int_{\underline{l}}^{\bar{l}} g(l)\phi_t(l)dl + \psi(t) + \varepsilon_t$$

- y_t is the oceangoing traffic at month t
- $g(l)$ is the potentially non-linear effect of lake level l on traffic
- $\phi_t(l)$ is the within-month distribution of daily lake levels
- $\psi(t)$ is the monthly trend capturing seasonal patterns
- ε_t is the error term.



Notes: Cubic spline baseline; shaded 90% CIs; Newey–West SEs (L=4).

Capacity Constraints and Rationing Mechanisms

	Transits (ships/mth)	Volume (Tons/mth)	Toll revenue (USD/mth)	Rev/Transit (USD/ship)	Slots available
Days below 82 ft	-0.0029***	-0.0044***	-0.0023***	0.0006*	-0.0040***
	(0.0010)	(0.0012)	(0.0008)	(0.0003)	(0.0014)
Days above 86 ft	0.0010	0.0022**	0.0009	-0.0001	0.0014*
	(0.0007)	(0.0009)	(0.0006)	(0.0004)	(0.0008)
Below 82 ft (%)					
+10 days	-2.88	-4.26	-2.30	0.60	-3.88
+30 days	-8.40	-12.24	-6.74	1.82	-11.20
Observations	267	267	267	267	267

Notes: Reference regime: 82–86 ft; Newey–West SEs (L=4); $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- +30 days below 82 ft reduces **Canal transits by 8.4%** and cargo volume by **12.2%**.
- The pattern is consistent with adjustment through **quantity restrictions**: slots available decline by 11.2%, while revenue per transit rises 1.8%. ▸ Detail

Days Below 82ft: $D_t^{<82} \equiv \sum_{d \in \mathcal{D}_t} \mathbf{1}\{l_{dt} < 82\}$, where l_{dt} is the daily Gatún Lake level.

Bilateral Trade Response to Canal Water Scarcity

$$\mathbb{E}[X_{ijt} \mid \cdot] = \exp\left[\underbrace{\left(\beta_L\, D_t^{<82} + \beta_H\, D_t^{>86}\right) \times \kappa_{ij}^{PC}}_{\text{Water regime} \times \text{PC dependence}} + \underbrace{\mu_{ij}}_{\text{Pair FE}} + \underbrace{\eta_{it} + \zeta_{jt}}_{\text{Exp./Imp.} \times \text{time FE}} + \underbrace{\psi_{PC}(t)}_{\text{Seasonal} \times \text{PC dep.}}\right]$$

Following [Heiland et al., 2025]: $\kappa_{ij}^{PC} \equiv \log(D_{ij}^{\text{NoPC}} / D_{ij}^{\text{Base}})$ measures the proportional increase in shortest-route distance when Panama is unavailable (e.g., New York-China: 17% farther, while New York-New Zeland: 59% farther without Panama).

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	Global	U.S. Trade	
	Value	Value	Weight
Days below 82 ft $\times \kappa_{ij}^{PC}$	−0.0008**	−0.0007	−0.0025*
	(0.0004)	(0.0007)	(0.0013)
Days above 86 ft $\times \kappa_{ij}^{PC}$	0.0001	0.0003	0.0013
	(0.0005)	(0.0007)	(0.0012)
Num. Obs.	4,115,373	1,428,476	1,428,476

Notes: PPML estimates. Driscoll–Kraay SEs (L=6). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- 30 days below 82 ft **reduces trade value by 0.4–1.2%**, depending on the detour.
- In U.S. data, the response is larger for **physical trade quantities** than for trade value.

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AIS Evidence on Maritime Rerouting

$$\mathbb{E}[V_{ijgt} \mid \cdot] = \exp\left[\underbrace{\sum_{h \neq PC} \mathbf{1}\{g = h\}\left(\beta_{Lh} D_t^{<82} + \beta_{Hh} D_t^{>86}\right)}_{\text{Water regime} \times \text{alternative corridor}} + \underbrace{\mu_{ijg}}_{\text{OD} \times \text{corridor FE}} + \underbrace{\lambda_{ijt}}_{\text{OD} \times \text{time FE}} + \underbrace{\psi_g(t)}_{\text{Season} \times \text{corridor}}\right]$$

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	Alternative	Cape	Magellan	Suez
Days below 82 ft	0.0322**	0.0297**	0.0397**	0.0193
	(0.0130)	(0.0122)	(0.0174)	(0.0307)
Days above 86 ft	−0.0246*	−0.0165	−0.0484**	0.0203
	(0.0142)	(0.0135)	(0.0202)	(0.0296)
Observations	30,669	30,669	30,669	30,669

Notes: PPML estimates. Voyages are assigned to departure month. Sample restricted to Origin-Destination pairs observed using Panama and at least one alternative corridor. "Alternative" pools Cape, Magellan, and Suez into a single category. Driscoll–Kraay SEs (L=6). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- Low-water conditions shift traffic **away from Panama toward alternative routes**.
- Reallocation is concentrated through **Cape and Magellan**.

Model: Trade on a Climate-Sensitive Network

- Countries connect through a **maritime network** $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ of ports and transit nodes
- Firms choose the **least-cost route** across all feasible paths [Allen and Arkolakis, 2022]:

$$X_{ij} = \tau_{ij}^{-\theta} \frac{Y_i}{\Pi_i} \frac{E_j}{P_j}, \quad \tau_{ij} = \left(\sum_{r \in \mathcal{R}_{ij}} \prod_{(l,l') \in r} t_{ll'}^{-\theta} \right)^{-1/\theta}$$

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- **Capacity at climate-sensitive nodes deteriorates under adverse local conditions**, raising baseline costs on affected links:

$$\bar{t}_{ll'}(\boldsymbol{\omega}) = \bar{t}^0_{ll'} \, \phi(\omega_l, \kappa_{ll'})$$

where ω_l measures climate stress at node l, $\kappa_{ll'} \in \{0, 1\}$ identifies exposed links, and

$$\phi(0, \kappa) = 1, \qquad \phi(\omega_l, 0) = 1, \qquad \partial \phi(\omega_l, 1)/\partial \omega_l > 0.$$

- Higher costs on constrained links $\Rightarrow$ **rerouting toward alternative paths**
- **Network spillovers** propagate to non-exposed routes and trading pairs

Calibration: Mapping Water Scarcity into Transport Costs

- **Equilibrium traffic on link (l, l'):**

$$\Xi_{ll',t} = \bar{t}_{ll',t}^{-\theta} \tilde{P}_{l,t}^{-\theta} \tilde{\Pi}_{l',t}^{-\theta}.$$

- **Panama transport costs respond to hydrological conditions:**

$$\bar{t}_{ll',t} = \bar{t}_{ll'}^{0} \exp\left[\mathbf{1}\{(l, l') \in \mathcal{E}_{PC}\} \left(\delta_L D_t^{<82} + \delta_H D_t^{>86}\right)\right]$$

- **Empirical analogue using AIS traffic:**

$$\mathbb{E}[\widehat{\Xi}_{ll't} \mid \cdot] = \exp\left[\mathbf{1}\{(l, l') \in \mathcal{E}_{PC}\} \left(\alpha_L D_t^{<82} + \alpha_H D_t^{>86}\right) + \mu_{ll'} + \eta_{lt} + \zeta_{l't} + \psi_{PC}(t)\right]$$

- **Mapping traffic into transport costs** [Heiland et al., 2025]:

$$\delta_L = -\alpha_L/\theta, \qquad \delta_H = -\alpha_H/\theta \qquad (\theta = 8).$$

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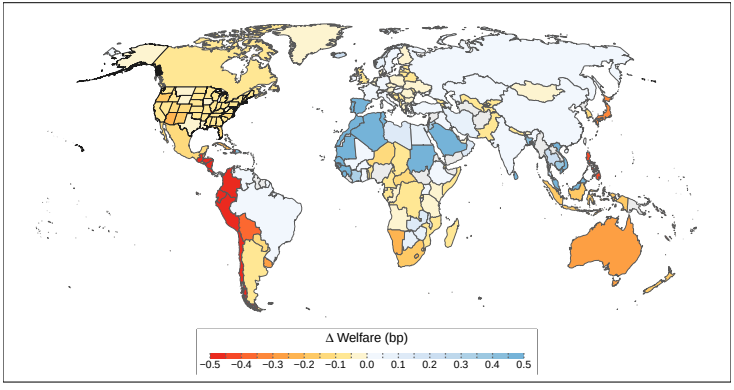
- **Mapping traffic into transport costs** [Heiland et al., 2025]:

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- **Calibrated low-water shock $\hat{\alpha}_L = -0.0109 \Rightarrow \hat{\delta}_L = 0.0014$** Detail:

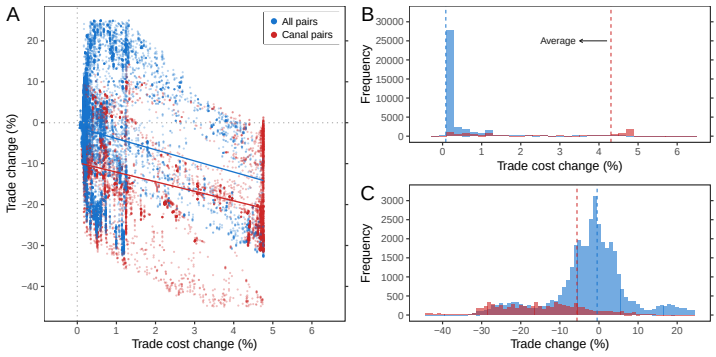
30 days below 82 ft implies **4.2% higher Panama transport costs**

Who Bears the Losses?



- *Counterfactual:* 30 days below 82 ft ▸ Method
- Deepest losses: Central America & Caribbean. All U.S. regions lose. ▸ Canal exposure
- Aggregate ≈ 0: **redistribution** with losses concentrated on Canal-dependent economies

Network Propagation in General Equilibrium



- Canal-dependent pairs face the largest **transport-cost increases and trade declines**.
- Endogenous rerouting **redistributes trade across the network**.
- The model implies a $\sim$**6% decline in trade** for Canal-dependent pairs.

Conclusions

Main finding: Water scarcity at a single reservoir can generate global economic effects.

- **Low water constrains Canal capacity.** Fewer slots limit how many ships can cross, and tighter draft limits constrain how much they can carry.
- **Alternatives shape exposure.** Trade falls more for pairs facing longer detours, while Cape and Magellan gain traffic relative to Panama.
- **The losses spread unevenly.** In the model, the shock propagates through trade links, with welfare losses concentrated in Central America and the Caribbean.

Next steps: Separate the effects of the direct cost shock, rerouting, and general equilibrium adjustment. Link precipitation projections [Muñoz et al., 2025] to Canal capacity and welfare.

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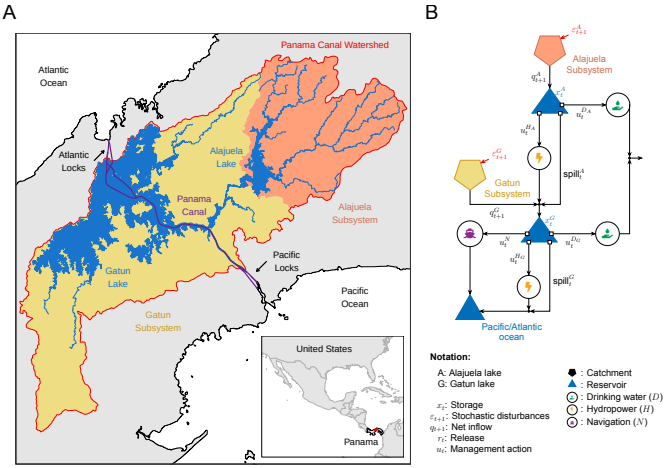
Which economies use the Panama Canal?

Routes	PCU/MS 2023	PCU/MS 2024	% Δ PCU/MS
East Coast U.S. - Asia	246,768,488	204,056,339	-17.31%
East Coast U.S. - W.C. South America	48,063,058	36,621,051	-23.81%
South America Intercoastal	26,037,602	30,509,715	17.18%
Europe - West Coast South America	25,799,446	20,293,355	-21.34%
East Coast U.S. - W.C. Central America	19,685,783	19,042,652	-3.27%
Europe - West Coast U.S.	15,457,079	14,181,845	-8.25%
U.S. Intercoastal including Alaska and Hawaii	11,884,991	13,580,741	14.27%
Central America Intercoastal	8,797,454	9,432,120	7.21%
E.C. Central America - W.C. South America	9,509,440	9,255,494	-2.67%
E.C. South America - Asia	16,336,445	7,944,913	-51.37%
E.C. South America - W.C. Central America	7,632,903	6,776,150	-11.22%
E.C. South America - West Coast U.S.	6,229,097	5,422,850	-12.94%
Asia - E.C. Central America	7,730,648	5,046,868	-34.72%
East Coast U.S. - Oceania	5,942,570	4,639,835	-21.92%
West Indies - Asia	5,164,159	3,979,433	-22.94%
W.C. Central America - Europe	3,016,240	2,960,520	-1.85%
Europe - Oceania	2,564,343	1,359,786	-46.97%
Asia - Europe	2,015,420	1,236,419	-38.65%
E.C. Central America - West Coast US	2,201,127	1,210,800	-44.99%
All Other Routes	40,300,417	25,542,186	-36.62%
Total	**511,136,710**	**423,093,072**	

Source: Panama Canal, Transit statistics. Panama Canal Universal Measurement System (PC/UMS)

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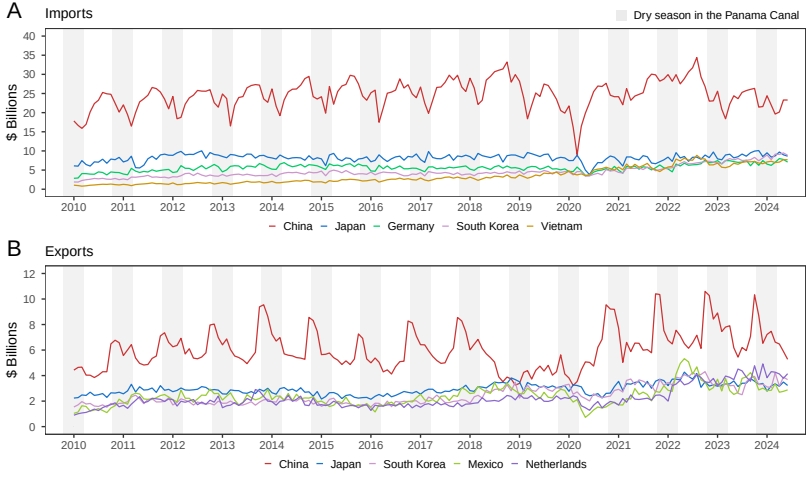
Schematic of the Panama Canal system



Source: [Prieto et al., 2026]

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US imports and exports



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AIS Traffic Calibration: Robustness

	(1) Baseline	(2) + PC controls	(3) + MR controls	(4) Preferred
Days below 82 ft × Panama	−0.0032**	−0.0036***	−0.0146	−0.0109**
	(0.0015)	(0.0010)	(0.0097)	(0.0050)
Days above 86 ft × Panama	0.0028**	0.0021**	−0.0007	0.0081
	(0.0014)	(0.0009)	(0.0048)	(0.0050)
OD × gate FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	–	–
Origin × time FE	–	–	Yes	Yes
Destination × time FE	–	–	Yes	Yes
Panama-specific year controls	–	Yes	–	Yes
Panama-specific month controls	–	Yes	–	Yes
Observations	2,941,286	2,941,286	2,941,286	2,941,286

Notes: PPML estimates on a common estimation sample. Driscoll–Kraay SEs (L=6). Column (4) is the preferred specification used to calibrate the Panama transport-cost shock. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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Hat Algebra: Solving the Counterfactual

- **Counterfactual cost shock:** $\hat{t}_{ll'} = \exp(\delta_R \Delta\omega\, \kappa^{PC}_{ll'})$
- **Fixed-point system** (solve for $\hat{p}_l$, $\hat{\pi}_l$):

$$\hat{p}_l = \sum_{l' \in \mathcal{V}} s_{l'l}\, \hat{t}_{l'l}^{-\theta}\, \hat{p}_{l'} + b_l^{\tilde{P}}(\hat{\boldsymbol{\pi}}) \qquad \hat{\pi}_l = \sum_{l' \in \mathcal{V}} s_{ll'}\, \hat{t}_{ll'}^{-\theta}\, \hat{\pi}_{l'} + b_l^{\tilde{\Pi}}(\hat{\mathbf{p}}, \hat{\boldsymbol{\pi}})$$

where $\hat{p}_l$ and $\hat{\pi}_l$ are the consumer-side and producer-side network potentials; $s_{ll'}$ is the baseline traffic shares; $b_l^{\tilde{P}}$, $b_l^{\tilde{\Pi}}$ capture local expenditure and income contributions; $b_l = 0$ for transit nodes $l \in \mathcal{V}_M$.

- **Recover counterfactual objects:**

Welfare

$$\hat{W}_i = \hat{\Pi}_i^{-\theta/(\theta+1)} / \hat{P}_i$$

Traffic

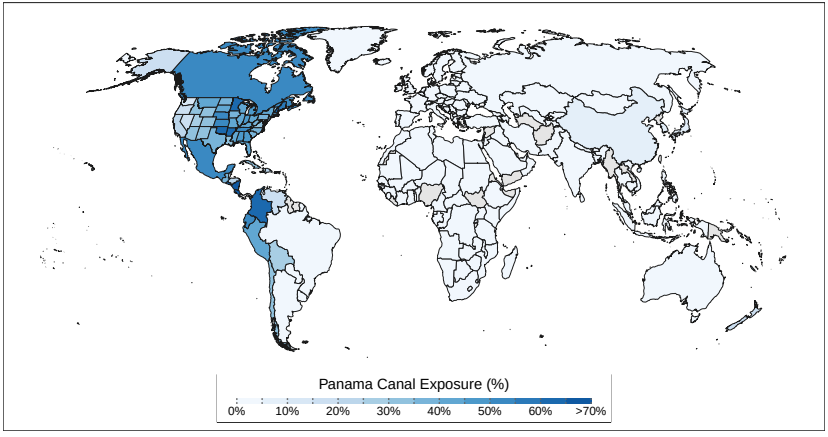
$$\hat{\Xi}_{ll'} = \hat{t}_{ll'}^{-\theta} \hat{P}_l^{-\theta} \hat{\Pi}_{l'}^{-\theta}$$

Trade flows

$$X'_{ij} = Y_i \left[\mathbf{D}^{X'} - \hat{\boldsymbol{\Xi}} \right]^{-1}_{ij} E_j$$

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Panama Canal Exposure



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Low-Water Effects: Rainfall Instruments

	Transits (ships/mth)	Volume (tons/mth)	Toll revenue (USD/mth)	Rev/Transit (USD/ship)	Slots available (slots/mth)
Days below 82 ft	−0.0040**	−0.0091***	−0.0042**	−0.0002	−0.0073**
	(0.0022)	(0.0030)	(0.0019)	(0.0013)	(0.0020)
Below 82 ft (% change)					
+10 days	-3.91	-8.67	-4.08	-0.19	-6.99
+30 days	-11.28	-23.82	-11.76	-0.57	-19.55
AR p-value	0.0176	0.0037	0.0405	0.9288	0.0111
HAC KP F	6.44	6.44	6.44	6.44	6.44
Observations	256	256	256	256	256

Notes: All dependent variables are in logs. Reference regime: lake levels at or above 82 ft; the separate high-water effect is fixed at zero. Instruments are accumulated rainfall deficits over preceding months 1–6 and 7–12. All models include month and year fixed effects. Parentheses report conventional Newey–West HAC standard errors ($L = 4$); brackets report 95% weak-IV-robust Anderson–Rubin confidence sets. Stars use HAC Anderson–Rubin tests under the maintained instrument restrictions: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Percentage effects are $100[\exp(d\hat{\beta}) - 1]$.

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Low-Water Effects: Inflow Instruments

	Transits (ships/mth)	Volume (tons/mth)	Toll revenue (USD/mth)	Rev/Transit (USD/ship)	Slots available (slots/mth)
Days below 82 ft	-0.0057^{*}	-0.0112^{**}	-0.0055^{***}	0.0003	-0.0082^{**}
	(0.0026)	(0.0039)	(0.0025)	(0.0017)	(0.0023)
Below 82 ft (% change)					
+10 days	-5.57	-10.58	-5.30	0.27	-7.90
+30 days	-15.79	-28.49	-15.08	0.81	-21.88
AR p-value	0.0547	0.0122	0.0005	0.2307	0.0154
HAC KP F	5.24	5.24	5.24	5.24	5.24
Observations	256	256	256	256	256

Notes: Dependent variables are in logs. Reference regime: lake levels at or above 82 ft; the separate high-water effect is fixed at zero. Instruments: accumulated Gatún inflow volumes over preceding months 1–6 and 7–12. Month and year fixed effects included. Parentheses report conventional Newey–West HAC standard errors ($L = 4$); brackets report 95% weak-IV-robust Anderson–Rubin confidence sets. Stars use HAC Anderson–Rubin tests under the maintained instrument restrictions: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Percentage effects are $100[\exp(d\hat{\beta}) - 1]$.

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